

the scientists' sweat and blood, they also have nature to thank, at least for that timely spark.

In another interesting example, the engineers of Energid Robotics and Machine Vision set out to develop a humanoid robot that can work like a human arm. The company's director and country manager, Jayakrishnan T., says, "Our unique robot manipulator series, the Cyton humanoid manipulators mimic nature's ultimate engineering marvel—the human! We copied the configuration of the human arm to do the kinematics design of that robot. These robots have the properties of redundancy and bifurcation, which enable placement of multiple hands or tools at desired positions and orientations within the robot's workspace in an unlimited number of ways. This was our main focus while we tried to design the robot following the human arm geometry. These humanoid arms can perform any manipulation task that a human being can do with his arm with a limited payload capacity of up to one kilogram." Cyton was completely designed and developed in India.

In general, robotics offers many examples of the nature-inspired kind. Energid itself has another example to offer: Their Citrus fruit harvester works much like how a frog hunts a bee.

Approach 4

"I was trying for long to solve a problem. Luckily, nature gave me a clue."

In the previous approach, the engineers decided consciously to study nature in order to find a solution to their problem. The experience of Dr Vinay Vaidya, chief technology officer, Engineering, and CREST Leader, KPIT Cummins Infosystems, is slightly different. Dr Vaidya was on the task of developing an image enhancement system for automobile, when he chanced upon a clue from nature in the form of the shape of the leaf. He was vigilant enough to notice it, research it further and use it to advance his technology.

Dr Vaidya recounts, "While we



Cyton humanoid manipulators mimic nature's ultimate engineering marvel—the human

were working on finding a solution to enhance an image, especially under night driving conditions, I came across a leaf with a shape similar to that of cardioids."

A cardioid has remarkable properties. The shape is formed in an interesting way. Take two circles of unit radius, rotate one circle over the circumference of another and the trajectory that is formed takes the shape of a cardioid.

Dr Vaidya continues, "Our image enhancement solution is solely inspired by the shape of the leaf, which helped us to enhance the image. We could devise a function around cardioids: If there are pixels which are very dark, then increase the pixel value; and re-

duce the value if there are brighter pixels. Through the entire exercise we actually came up with a solution for better images under night driving conditions. We have filed for a patent that includes cardioid method along with other novelties for night vision," he says.

Inspiration or imitation?

Nature-inspired technologies fall into a certain classification, or rather progression, of terminologies such as bio-inspired, biomimetic, biomorphic and bio-replication.

Bio-inspired. Any artificial design, process or product that is inspired by nature is a bio-inspired one. Some researchers say that it is an umbrella term that covers all the others, while some restrict it to just those products which exhibit a natural functionality but do not employ the underlying natural principles.

Bio-morphic. "Bio-inspired just takes inspiration from biology, whereas bio-morphic seeks to exploit the same solutions or employ the same organising principles. The term 'morphic' comes from structure, emphasising arriving at the same function by mimicking the underlying structure or architecture," explains Dr Kwabena Boahen, associate professor, Bioengineering Department, Stanford University.

Dr Boahen is principal investigator of their Brain in Silicon project—a biomorphic or rather neuromorphic one. The team tries to use existing knowledge of the brain's functioning to design an affordable supercomputer. The other complementary side of their research is to use the same supercomputer to further investigate the functioning of the brain.

According to the team, "We model brains using an approach far more efficient than software simulation: We emulate the flow of ions directly with the flow of electrons—don't worry, on the outside it looks just like software."

Developed through a neuromorphic approach, this supercomputer, called the Neurogrid, has a parallel,

Strongest playgrounds

Some areas of science and technology where biomimetic techniques have been found to be very effective are:

1. Machine vision systems
2. Machine hearing systems
3. Signal amplifiers
4. Navigational systems
5. Data converters
6. Neural networks
7. Nanorobot antibodies (to seek and destroy disease-causing bacteria)
8. Artificial organs, arms, legs, hands and feet
9. Implantable devices